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Đoàn Văn Giáp

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RESEARCH ON INTRINSIC MOTIVATION FOR REINFORCEMENT LEARNING

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Doan Van Giap

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Declaration of Authorship

I hereby declare that this thesis is my own work and has not been submitted previously for a degree or diploma at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where proper acknowledgment and citation are given.

Hanoi, ...

Student

Doan Van Giap

Tóm tắt

Khóa luận *Research on Intrinsic Motivation for Reinforcement Learning* nghiên cứu bài toán học tăng cường trong môi trường phần thưởng thưa, nơi tác tử rất khó khám phá do tín hiệu phản hồi từ môi trường xuất hiện không thường xuyên. Nhiều phương pháp hiện nay kết hợp phần thưởng ngoại sinh với phần thưởng nội tại dựa trên tò mò hoặc độ mới, nhưng thường phải dùng một hệ số cố định để cân bằng hai nguồn phần thưởng này. Hệ số cố định dễ nhạy với từng bài toán và không phản ánh được việc ở những trạng thái khác nhau thì nhu cầu khám phá cũng khác nhau.

Khóa luận đề xuất **Correlation-Aware Reward Exploration (CARE)**, một cơ chế điều chỉnh phần thưởng nội tại theo từng trạng thái thông qua một Beta Network gọn nhẹ. Mạng này được huấn luyện bằng mục tiêu tương quan nhằm làm cho phần thưởng nội tại sau khi được scale có quan hệ chặt chẽ hơn với tổng phần thưởng ngoại sinh trong tương lai. Phương pháp được kết hợp với PPO và Intrinsic Curiosity Module để vừa tăng cường khả năng khám phá, vừa giữ được tính ổn định trong quá trình huấn luyện.

Kết quả thực nghiệm trên sáu môi trường MiniGrid (DoorKey-8x8, Empty-16x16, KeyCorridor-S3R3, LavaCrossing-S9N3, RedBlueDoors-8x8, UnlockPickup) cho thấy CARE cải thiện hiệu quả mẫu và giảm độ dao động so với PPO thuần túy cũng như PPO kết hợp ICM với hệ số cố định. Phương pháp đặc biệt hiệu quả trong các môi trường có phần thưởng thưa nhưng vẫn còn tín hiệu ngoại sinh mang tính định hướng. Trong những bài toán quá thưa, CARE suy giảm một cách ổn định về gần hành vi của hệ số cố định thay vì gây mất ổn định cho quá trình học.

Từ khóa: học tăng cường, phần thưởng thưa, phần thưởng nội tại, khám phá dựa trên tò mò, điều chỉnh phần thưởng thích ứng

Abstract

Sparse-reward reinforcement learning remains difficult because agents receive limited feedback about whether their behavior is useful. A common solution is to augment the environment reward with intrinsic motivation signals such as novelty, curiosity, or impact. However, count-based bonuses, the Intrinsic Curiosity Module (ICM), and Random Network Distillation (RND) are all commonly combined with extrinsic rewards through a fixed coefficient. This design makes performance highly sensitive to manual tuning and obscures a broader empirical question: how does an adaptive scaling mechanism affect different intrinsic reward modules?

This thesis proposes **Correlation-Aware Reward Exploration (CARE)**, a module-agnostic adaptive wrapper for intrinsic reward modules. CARE learns a state-dependent scaling factor through a lightweight Beta Network and uses a correlation-based objective to align weighted intrinsic rewards with discounted future extrinsic returns. Under this framing, CARE is not treated only as a standalone method paired with one curiosity bonus, but as a mechanism for studying and improving how representative intrinsic reward modules are used during training.

The thesis therefore analyzes CARE with respect to three representative intrinsic reward families: count-based exploration, ICM, and RND. In the implemented system, the quantitative experiments instantiate the framework with Proximal Policy Optimization and ICM on six sparse-reward MiniGrid environments (DoorKey-8x8, Empty-16x16, KeyCorridor-S3R3, LavaCrossing-S9N3, RedBlueDoors-8x8, and UnlockPickup), while count-based and RND modules are used to motivate the comparative experimental framing and discussion. The results show that adaptive scaling improves sample efficiency and reduces variance relative to PPO without intrinsic rewards and PPO with fixed intrinsic coefficients. Overall, the study suggests that CARE is best understood as a practical adaptive layer whose benefits depend on the structure and informativeness of the underlying intrinsic reward module.

Keywords: reinforcement learning, sparse rewards, intrinsic motivation, adaptive reward scaling, count-based exploration, ICM, RND

Contents

List of Figures

List of Tables

List of Acronyms

Acronym	Full Form	Description
CARE	Correlation-Aware Reward Exploration	Proposed adaptive intrinsic reward scaling framework
RL	Reinforcement Learning	Learning paradigm based on sequential interaction with an environment
PPO	Proximal Policy Optimization	On-policy policy-gradient algorithm used as the training backbone
ICM	Intrinsic Curiosity Module	Intrinsic reward model based on forward prediction error
RND	Random Network Distillation	Curiosity-based method using predictor error against a random target network
RIDE	Rewarding Impact-Driven Exploration	Intrinsic reward method based on state-change impact
GAE	Generalized Advantage Estimation	Technique for reducing variance in policy-gradient advantage estimates
MDP	Markov Decision Process	Standard formal model for reinforcement learning environments

List of Notations

Symbol	**Meaning**
	Markov decision process and policy
$\mathcal{S}, \mathcal{A}$	State space and action space
$P(s' \mid s, a)$	State transition kernel
ρ_0	Initial state distribution
$\gamma \in [0, 1)$	Discount factor
$\pi_\theta(a \mid s)$	Stochastic policy with parameters θ
$V_\mu(s)$	State-value function with parameters μ
$\hat{A}_t$	Generalized advantage estimate at time t
$\rho_t(\theta)$	PPO importance ratio $\pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$
	Reward signals
R_t^E	Extrinsic reward from the environment at time t
R_t^I	Generic intrinsic reward at time t (any module)
$R_t^{I,m}$	Intrinsic reward at time t produced by module m
I_t	Raw intrinsic reward (forward prediction error in the ICM case)
$I_t^+,\ I_t^{+,m}$	Standardized and rectified intrinsic reward (generic / module-specific)
$\tilde{I}_t^m$	Scaled intrinsic reward $\beta_\psi(s_t)\, I_t^{+,m}$
$\bar{r}_t$	Shaped reward used by PPO
G_t^E	Discounted extrinsic return from time t
$\hat{I}_t^m,\ \hat{G}_t$	Within-batch standardized scaled intrinsic and extrinsic return
	ICM components
$\phi(s_t)$	ICM state encoder
f_η^F	ICM forward dynamics model with parameters η
f_η^I	ICM inverse dynamics model with parameters η
α_F, α_I	Forward and inverse loss weights in L_{ICM}
	CARE components
α	Global intrinsic-strength hyperparameter
$\beta_\psi(s)$	State-dependent scaling function with parameters ψ

Chapter 1

Introduction

> *"The important thing is not to stop questioning. Curiosity has its own reason for existing."*
>
> — Albert Einstein

Curiosity is widely viewed as a driving force behind learning, both in humans and in artificial systems. In reinforcement learning (RL), this idea has been formalized as intrinsic motivation: an internally generated signal that encourages an agent to seek out novel or informative experiences when external feedback is scarce. Yet the question of *how strongly* an agent should listen to its own curiosity remains largely unresolved. Most existing methods rely on a single, manually chosen coefficient that controls the influence of intrinsic rewards uniformly across all states. This thesis takes the position that the importance of curiosity is itself state-dependent and proposes a learned scaling mechanism that adapts the influence of intrinsic motivation to the current situation of the agent.

1.1 Background and Research Motivation

Reinforcement learning has emerged as a central paradigm for training agents to make sequential decisions in complex environments, with influential successes in Atari games, Go, and robotic control [1–3]. Many of these results, however, depend on dense reward signals that provide frequent feedback about which behaviors are desirable. In contrast, a wide range of practical environments are sparse-reward: the agent receives meaningful feedback only after a long sequence of actions, which makes exploration substantially harder [4].

To address this difficulty, a large body of research augments the extrinsic reward supplied by the environment with an additional intrinsic reward that encourages exploration [5, 6]. Count-based methods reward the agent for visiting rarely seen states [7–9]. Curiosity-driven methods derive intrinsic rewards from prediction errors or representation discrepancies, as in the Intrinsic Curiosity Module (ICM)

[10], Random Network Distillation (RND) [11], and Rewarding Impact-Driven Exploration (RIDE) [12]. These approaches improve exploration in many tasks, but they all share a key design choice: the strength with which intrinsic rewards influence learning relative to extrinsic rewards.

In standard practice, this strength is controlled by a fixed scalar coefficient that must be tuned manually for each environment. The optimal value of this coefficient often varies across tasks and even across different stages of training [13, 14]. More fundamentally, a single global coefficient cannot distinguish between states where exploration remains useful and states where exploration has become distracting or harmful. The same limitation appears across multiple intrinsic reward families, from count-based bonuses to ICM and RND. This shared weakness motivates a broader research perspective: rather than studying yet another curiosity module, this thesis investigates an adaptive mechanism whose effect can be analyzed across representative intrinsic reward modules.

1.2 Research Problem

This thesis investigates the following problem: how does adaptive, state-dependent intrinsic reward scaling affect different intrinsic reward modules, and can such scaling promote useful exploration in sparse-reward reinforcement learning without destabilizing policy optimization?

More specifically, let the agent receive an extrinsic reward R_t^E from the environment and an intrinsic reward $R_t^{I,m}$ produced by intrinsic reward module m, where m may denote a count-based bonus, ICM, or RND. Standard methods optimize a shaped reward of the form

$$\bar{r}_t = R_t^E + \beta R_t^{I,m}, \tag{1.1}$$

where β is a fixed hyperparameter. The limitation of this formulation is that the same intrinsic scaling is applied uniformly to all states, even though the value of exploration may differ substantially across the state space. The central research question of this thesis is therefore:

> *Can a state-dependent scaling mechanism, learned online from the agent's own experience, improve how representative intrinsic reward modules such*

as count-based bonuses, ICM, and RND contribute to learning in sparse-reward environments?

1.3 Research Objectives and Scope

The objective of this thesis is to develop and empirically evaluate **Correlation-Aware Reward Exploration (CARE)**, a state-dependent scaling mechanism for intrinsic rewards, and to study its effect on representative intrinsic reward modules in sparse-reward reinforcement learning. The specific objectives are as follows:

- analyze why fixed intrinsic reward coefficients are a shared limitation across representative intrinsic reward modules;
- review count-based exploration, ICM, RND, and prior adaptive reward weighting approaches as the foundation for state-dependent adaptation;
- formulate CARE as a lightweight, module-agnostic scaling layer that learns a state-dependent coefficient $\beta(s_t)$ via a correlation-based objective;
- describe the concrete implementation of CARE with Proximal Policy Optimization (PPO) and ICM, while clarifying how the same formulation extends to count-based and RND modules;
- evaluate the empirical effect of adaptive scaling in sparse-reward MiniGrid environments and discuss what the results imply for different intrinsic reward modules.

The scope of this thesis is limited to episodic reinforcement learning tasks with sparse rewards. Conceptually, the study is framed around three representative intrinsic reward families: count-based bonuses, ICM, and RND. In implementation, the quantitative experiments use PPO as the base policy optimization algorithm and ICM as the instantiated intrinsic reward generator. Count-based and RND modules are included in the literature review and experimental framing to position CARE as a general adaptive layer rather than an ICM-specific technique. The experiments focus on benchmark MiniGrid environments [15] rather than real-world robotics or continuous-control tasks. The thesis emphasizes empirical performance and methodological clarity rather than a full theoretical proof of convergence or optimality.

1.4 Research Methodology

The work follows a research-oriented methodology with four stages. First, the thesis surveys foundational RL literature and representative intrinsic reward modules to identify the common limitation of fixed intrinsic weighting. Second, it formulates CARE as an adaptive scaling mechanism that predicts a positive coefficient $\beta(s_t)$ from the current state and can be applied to module-specific intrinsic rewards. Third, the method is instantiated within a PPO training loop using ICM-derived intrinsic rewards as the primary quantitative case study. Finally, the approach is validated experimentally on multiple sparse-reward tasks in MiniGrid by comparing it against PPO without intrinsic rewards and PPO with fixed intrinsic coefficients.

The evaluation criteria emphasize sample efficiency, learning stability, robustness across environments, and the qualitative behavior of the learned scaling function. This combination of theoretical grounding, algorithmic design, and empirical validation supports an experimental framing centered on the effect of CARE on intrinsic reward modules rather than on the construction of a new intrinsic reward generator.

1.5 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews reinforcement learning in sparse-reward environments, representative intrinsic reward modules, and prior attempts to adapt the weighting of intrinsic rewards; it also identifies the research gap that motivates the proposed method. Chapter 3 presents the proposed methodology, including the formal problem statement, the module-agnostic CARE formulation, the correlation-based training objective, the concrete ICM instantiation, and the complete training algorithm; it also states five formal results (Propositions, a Lemma, and a Corollary) that characterize CARE's properties. Chapter 4 describes the experimental setup, including the MiniGrid benchmark environments, the implementation details, the evaluation metrics, and the analysis of results from the perspective of CARE's effect on intrinsic reward modules. The conclusion summarizes the main findings, discusses limitations of the proposed approach, and outlines directions for future research. Two appendices complete the thesis: Appendix A contains the full proofs of the formal results introduced in Chapter 3, and Appendix B records the network architectures, hyperparameters, and training schedule used in the experiments.

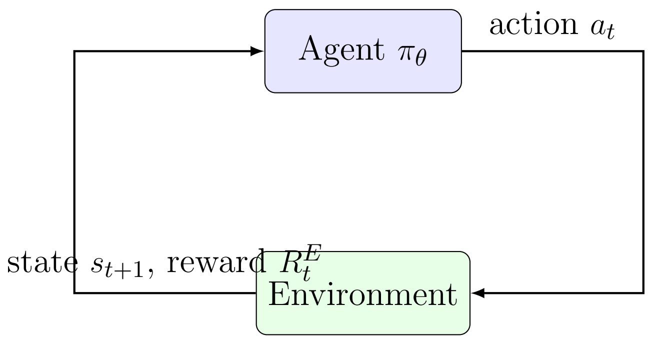
Chapter 2

Literature Review

This chapter summarizes the published studies most relevant to this thesis. It first formalizes reinforcement learning in sparse-reward environments and discusses why standard exploration is insufficient in such settings. It then reviews representative intrinsic motivation methods, including count-based and curiosity-driven exploration, and surveys prior attempts to adapt the weighting of intrinsic rewards. The chapter concludes by identifying the specific gap that motivates the proposed methodology in Chapter 3.

2.1 Reinforcement Learning in Sparse-Reward Environments

An episodic reinforcement learning (RL) problem is commonly modeled as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \rho_0)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s_{t+1} \mid s_t, a_t)$ denotes the transition dynamics, R is the reward function, $\gamma \in [0, 1)$ is the discount factor, and ρ_0 is the initial state distribution [3]. At each time step t, the agent observes a state $s_t \in \mathcal{S}$, samples an action $a_t \sim \pi_\theta(\cdot \mid s_t)$ from a parameterized policy, receives an extrinsic reward R_t^E, and transitions to a new state s_{t+1}. This standard interaction loop is depicted in Figure 2.1.



Hình 2.1: Standard agent–environment interaction loop in reinforcement learning. The agent observes a state, executes an action sampled from its policy, and receives a new state along with an extrinsic reward from the environment.

The objective of policy optimization is to learn parameters θ that maximize the expected discounted return:

$$J(\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, P)} \left[\sum_{t=0}^{T-1} \gamma^t R_t^E \right], \tag{2.1}$$

where $\tau = (s_0, a_0, s_1, a_1, \dots)$ denotes a trajectory induced by the policy and the transition dynamics.

Although Equation (2.1) is a standard formulation, learning becomes difficult when extrinsic rewards are sparse. In sparse-reward tasks, the agent may take a long sequence of actions before receiving any feedback that distinguishes promising trajectories from uninformative ones. Classical exploration strategies such as ϵ-greedy or stochastic policies inject randomness into action selection, but they do not actively guide the agent toward informative experiences when the state space is large or partially observable [4]. As a result, an agent relying solely on extrinsic rewards in such environments often fails to discover the goal-reaching behaviors needed for learning to begin.

2.2 Intrinsic Motivation for Exploration

Intrinsic motivation addresses sparse rewards by supplementing the environment's reward with an additional exploration bonus that is internally generated by the agent itself [5, 6]. Under this paradigm, the agent receives both an extrinsic reward R_t^E and an intrinsic reward R_t^I, and policy optimization is performed on a shaped objective:

$$\bar{r}_t = R_t^E + \beta R_t^I, \tag{2.2}$$

where $\beta \geq 0$ controls the relative importance of intrinsic motivation. The intrinsic signal R_t^I is designed to be larger for states or transitions that the agent finds novel, surprising, or otherwise informative. A recent survey by Aubret *et al.* [16] unifies many of these methods through an information-theoretic perspective, showing that seemingly different intrinsic reward generators can be understood as different practical instantiations of similar underlying principles.

2.2.1 Count-Based Exploration

Count-based methods assign higher intrinsic rewards to states that have been visited less frequently. In tabular settings, the intrinsic reward at state s is often inversely proportional to the square root of its visit count $N(s)$, capturing the intuition that under-explored states are more valuable to revisit. Bellemare *et al.* [7] extend this principle to high-dimensional observations through pseudo-counts derived from density models, while Tang *et al.* [8] use random projection hashing to obtain approximate counts in continuous state spaces. Ostrovski *et al.* [9] further refine the approach with neural density models, demonstrating strong exploration on hard Atari games such as *Montezuma's Revenge*. Count-based methods are conceptually simple and often effective, but the quality of the intrinsic signal is bounded by the ability of the underlying density estimator to reliably quantify novelty in complex observation spaces.

2.2.2 Curiosity-Driven Exploration

Curiosity-driven methods define intrinsic rewards through some form of prediction error. The Intrinsic Curiosity Module (ICM) [10] learns a compact latent representation of the state and trains a forward model that predicts the next latent feature from the current feature and action. The prediction error of this forward model is then used as an intrinsic bonus, encouraging the agent to seek transitions whose dynamics it cannot yet predict accurately. Random Network Distillation (RND) [11] replaces the forward dynamics model with the discrepancy between a fixed random target network and a learned predictor, providing a measure of state novelty that does not depend on action prediction. Rewarding Impact-Driven Exploration (RIDE) [12] computes intrinsic rewards from the magnitude of representation changes that are causally induced by the agent's actions, which makes it particularly suited to procedurally generated environments where stochastic background changes might otherwise dominate prediction-based bonuses.

A common advantage of curiosity-driven methods is that intrinsic rewards tend to decrease naturally as the agent becomes familiar with a region of the state space, reducing the risk of repeatedly exploring the same area. However, this self-correcting property only operates at the level of the raw intrinsic signal; the relative weight of that signal in the shaped reward of Equation (2.2) is still controlled by an external coefficient β that does not adapt to context.

2.3 Adaptive Weighting of Intrinsic Rewards

Despite the diversity of intrinsic motivation modules, most of them depend on a manually selected coefficient β in the shaped reward of Equation (2.2) [10, 11]. This fixed coefficient creates a structural limitation. If β is too small, the intrinsic signal becomes negligible and exploration remains weak. If β is too large, curiosity may dominate the training objective and distract the agent from the underlying task. Because the optimal trade-off varies across environments and even across different stages of the same training run, fixed weighting lacks flexibility. Several lines of research have attempted to overcome this limitation, with adaptation operating at progressively finer granularity.

2.3.1 Global-Level Adaptation

A first family of approaches treats the trade-off between extrinsic and intrinsic rewards as a global optimization problem. Extrinsic-Intrinsic Policy Optimization (EIPO) [13] casts intrinsic reward weighting as a constrained optimization in which the influence of intrinsic motivation is increased only when the extrinsic objective allows for it, preventing intrinsic rewards from harming task performance. Automatic Intrinsic Reward Shaping (AIRS) [14] frames the selection among different intrinsic reward functions as a bandit-style decision process, automatically choosing which exploration signal is most beneficial at each phase of training. Both methods improve over hand-tuned coefficients, but their adaptation is global: a single weight or a single intrinsic reward type is selected for the entire policy at any given time, regardless of the specific state being visited.

2.3.2 Population-Level Adaptation

A second family of approaches coordinates a population of policies with different exploration–exploitation trade-offs. Never Give Up (NGU) [17] maintains multiple value functions associated with different intrinsic reward weights, allowing the agent to share a single neural network across distinct exploratory regimes. Agent57 [18] extends this idea by additionally adapting the discount factor and weighting through a meta-controller that selects among the population members based on their recent performance. These methods substantially improve robustness across diverse Atari benchmarks, but their adaptation occurs at the level of an entire policy, not at the level of individual states. The shaping of intrinsic rewards within any single policy of the population is still governed by a fixed coefficient.

2.3.3 Toward State-Level Adaptation

A third, more direct line of work attempts to learn the intrinsic reward signal itself. Zheng *et al.* [19] propose Learning Intrinsic Rewards for Policy Gradient methods (LIRPG), in which a parameterized intrinsic reward function is trained via meta-gradients so that, after a policy update using the shaped reward, the resulting policy maximizes the extrinsic objective. This formulation is conceptually appealing because the intrinsic reward is directly aligned with future extrinsic performance. In practice, however, meta-gradient training requires differentiating through one or more policy updates, which introduces substantial computational overhead and can be unstable in deep RL settings.

The conceptual gap that emerges from this body of work is clear. Existing adaptive methods either operate at a coarse granularity (global or population-level) or rely on computationally expensive higher-order optimization (LIRPG-style meta-gradients). What is missing is a lightweight mechanism that adapts the influence of intrinsic rewards at the state level, learned via a first-order objective, and that can be plugged on top of different intrinsic reward generators without modifying their internal mechanics.

2.4 Summary and Research Gap

This chapter has reviewed the foundations on which the proposed work is built. Reinforcement learning in sparse-reward environments is challenging because conventional exploration strategies provide only weak guidance. Intrinsic motivation methods such as count-based bonuses, ICM, RND, and RIDE alleviate this difficulty by injecting an internally generated exploration signal, but the strength of that signal is typically governed by a fixed scalar coefficient. Prior work on adaptive weighting addresses this limitation only partially: global and population-level adaptation modulate intrinsic motivation at coarse granularity, while meta-gradient methods such as LIRPG operate at the right level but at a high computational cost.

Two related research gaps emerge from this analysis. First, there is no lightweight, state-dependent mechanism that modulates intrinsic rewards according to how well they correlate with future extrinsic outcomes, despite the fact that two states with similar novelty scores can contribute very differently to downstream task success. Second, although count-based bonuses, ICM, and RIDE share the same fixed-coefficient pattern, the literature does not provide a unified experimental framing that asks whether a single adaptive mechanism can improve them in a consistent way.

The methodology presented in Chapter 3 is designed to address both gaps. It introduces a state-dependent scaling function that is trained with a first-order correlation-based objective and that operates as a module-agnostic adaptive layer above representative intrinsic reward generators. The remainder of the thesis develops this approach in detail and evaluates its empirical effect on intrinsic reward modules in sparse-reward MiniGrid benchmarks.

Chapter 3

Methodology

This chapter presents the proposed methodology in formal terms. It first defines the underlying intrinsic-augmented decision process and the overall system architecture. It then specifies the intrinsic reward generator used in the implemented system, formulates the Correlation-Aware Reward Exploration (CARE) framework with its Beta Network and correlation-based objective, and concludes with the complete training algorithm. Five formal results (Propositions 3.1–3.3, Lemma 3.1, Corollary 3.1) are stated in this chapter and proved in Appendix A.

3.1 System Architecture

The proposed framework is built on top of an intrinsic-augmented Markov decision process (MDP) and introduces a learned, state-dependent scaling function for intrinsic rewards. The two ingredients are made explicit in the following definitions.

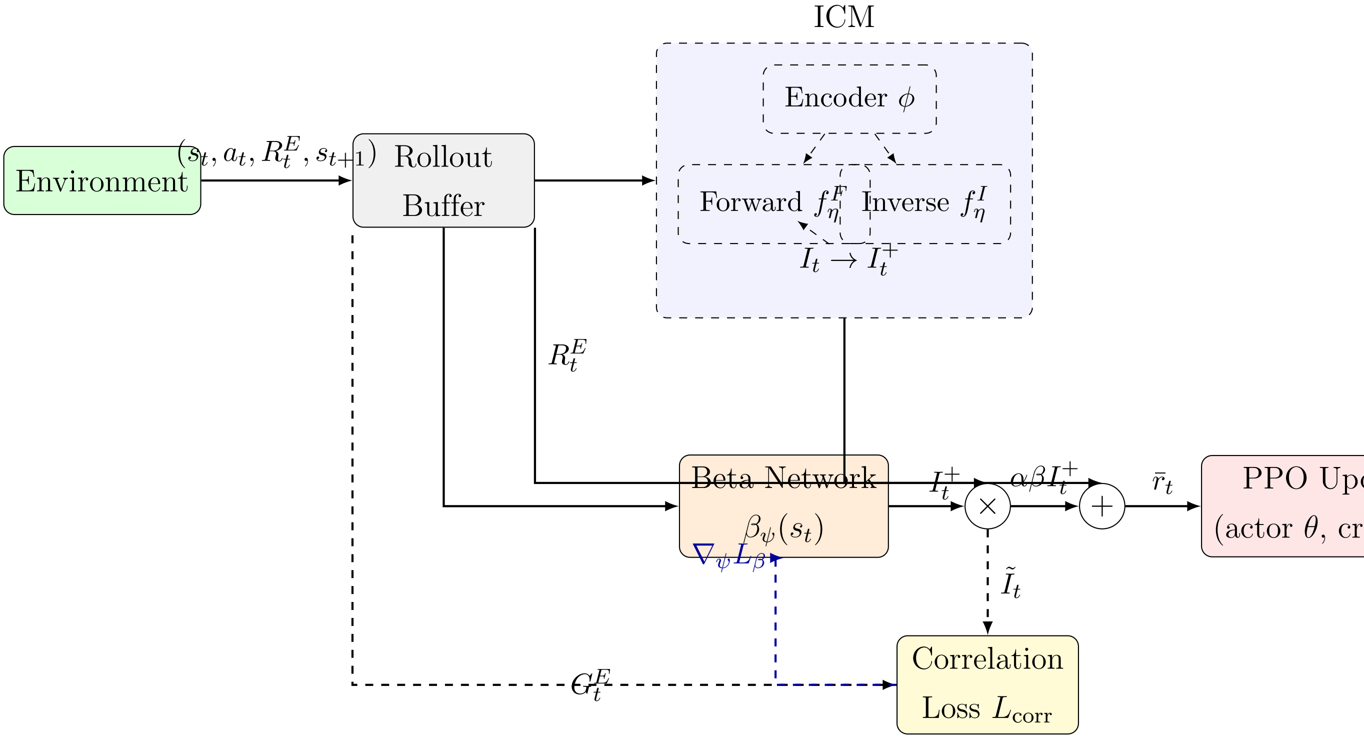
Definition 3.1 (Intrinsic-Augmented MDP)**.** An *intrinsic-augmented MDP* is a tuple $(\mathcal{S}, \mathcal{A}, P, R^E, R^I, \gamma, \rho_0)$ in which $(\mathcal{S}, \mathcal{A}, P, R^E, \gamma, \rho_0)$ defines a standard MDP with extrinsic reward function $R^E : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, and $R^I : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}_{\geq 0}$ is an additional, internally generated intrinsic reward function. The agent observes only $(s_t, a_t, R^E_t, s_{t+1})$ from the environment; the intrinsic reward R^I_t is computed by the agent itself.

Definition 3.2 (State-Dependent Reward Shaping)**.** A *state-dependent reward shaping function* is a measurable map $\beta : \mathcal{S} \to \mathbb{R}_+$. Given an intrinsic-augmented MDP and a global intrinsic-strength hyperparameter $\alpha > 0$, the corresponding shaped reward at time step t is

$$\bar{r}_t = R^E_t + \alpha\, \beta(s_t)\, R^I_t. \tag{3.1}$$

The standard fixed-coefficient formulation corresponds to the special case $\beta(s) \equiv 1$.

The proposed system instantiates this formulation with a parameterized β_ψ and a specific intrinsic reward generator (ICM in the implemented experiments). The full pipeline is shown in Figure 3.1.



Hình 3.1: CARE system architecture. Solid arrows denote the forward computation that produces the shaped reward $\bar{r}_t$ used by PPO. Dashed arrows indicate the auxiliary signals used to update the Beta Network β_ψ via the correlation-based objective L_β.

The shaped reward used throughout this thesis follows directly from Definition 3.2. For the chosen intrinsic reward module m with rectified intrinsic signal $I_t^{+,m}$ (Section 3.2), the combined reward optimized by PPO is

$$\bar{r}_t \;=\; R_t^E \;+\; \alpha\, \beta_\psi(s_t)\, I_t^{+,m}. \tag{3.2}$$

Conceptually, $\beta_\psi(s_t)$ acts as a state-conditional volume control on curiosity: it amplifies intrinsic reward in states where exploration is empirically associated with future extrinsic success and attenuates it elsewhere. The remainder of this chapter formalizes this intuition.

3.2 Intrinsic Reward Generation: ICM Instantiation

In the implemented system, the intrinsic reward generator is the Intrinsic Curiosity Module (ICM) [10]. Although CARE is module-agnostic, ICM provides a wellstudied and computationally manageable case for quantitative evaluation.

Forward and Inverse Dynamics Models. ICM first encodes each state into a latent representation $\phi(s_t) \in \mathbb{R}^d$. It then trains two auxiliary models: a forward dynamics model f_η^F that predicts the next latent feature from the current feature and action, and an inverse dynamics model f_η^I that predicts the action from two consecutive latent states. The combined ICM loss is

$$L_{\text{ICM}}(\eta) \;=\; \alpha_F \, \mathbb{E}\Big[\tfrac{1}{2}\big\|f_\eta^F(\phi_t, a_t) - \phi_{t+1}\big\|_2^2\Big] \;+\; \alpha_I \, \mathbb{E}[-\log p_\eta(a_t \mid \phi_t, \phi_{t+1})]\,, \tag{3.3}$$

with weighting coefficients $\alpha_F, \alpha_I > 0$.

Raw Intrinsic Reward. The raw intrinsic reward at time step t is the forward prediction error

$$I_t \;=\; \tfrac{1}{2}\big\|f_\eta^F(\phi_t, a_t) - \phi_{t+1}\big\|_2^2\,. \tag{3.4}$$

Because the magnitude of I_t is not stationary across training, the raw signal is preprocessed before being passed to CARE.

Standardization and Rectification.

Definition 3.3 (Standardized Intrinsic Reward)**.** Given a minibatch with empirical mean μ_I and standard deviation σ_I of the raw intrinsic rewards, and a small constant $\varepsilon > 0$, the *standardized intrinsic reward* is

$$I_t^+ \;=\; \max\!\left(0,\; \frac{I_t - \mu_I}{\sigma_I + \varepsilon}\right). \tag{3.5}$$

This preprocessing suppresses transitions whose prediction errors are below the batch average and retains relatively surprising transitions, which keeps the input to the Beta Network bounded in scale and approximately stationary across updates. In what follows, we use the generic notation $I_t^{+,m}$ to emphasize that the same construction applies to other intrinsic reward modules: replacing ICM with a count-based bonus or with Random Network Distillation (RND) requires changing the definition of I_t but leaves the rest of the pipeline (standardization, Beta Network, correlation loss) unchanged.

3.3 Correlation-Aware Reward Exploration

CARE introduces a state-dependent scaling function $\beta_\psi(s_t)$ between the intrinsic reward generator and the policy update. This section formalizes the network, its training objective, and the reasons for choosing a first-order optimization scheme over a meta-gradient one.

3.3.1 Beta Network

Definition 3.4 (Beta Network)**.** The *Beta Network* is a parameterized map

$$\beta_\psi : \mathcal{S} \to [\beta_{\min}, \beta_{\max}], \qquad 0 < \beta_{\min} < \beta_{\max} < \infty, \tag{3.6}$$

which assigns a positive, bounded scaling factor to each state. In the implemented system, β_ψ is realized as a two-layer multilayer perceptron with embedding dimension 256, sigmoid output, and clipping to $[\beta_{\min}, \beta_{\max}] = [0.1, 2.0]$.

For an intrinsic reward module m, the scaled intrinsic reward at time step t is

$$\tilde{I}_t^m \;=\; \beta_\psi(s_t)\, I_t^{+,m}, \tag{3.7}$$

and substituting Equation (3.7) into Equation (3.2) yields the shaped reward used by PPO.

The boundedness assumption built into Definition 3.4 ensures that the shaped reward cannot grow arbitrarily large relative to the extrinsic signal, regardless of the behavior of the Beta Network during training.

Lemma 3.1 (Bounded Shaped Reward)**.** *Let β_ψ satisfy Definition 3.4 and assume the standardized intrinsic reward is bounded almost surely, $|I_t^{+,m}| \leq M$ for some constant $M < \infty$. Then for every time step t,*

$$\left|\bar{r}_t - R_t^E\right| \;\leq\; \alpha\, \beta_{max}\, M.$$

Proof in Appendix A.3.

Lemma 3.1 is what allows CARE to be inserted into a PPO training loop without introducing additional instability: the shaped reward differs from the extrinsic reward by a bounded perturbation, so the policy gradient remains bounded under the same regularity conditions that PPO already requires.

3.3.2 Correlation-Based Objective

The Beta Network is trained so that the scaled intrinsic reward $\tilde{I}_t^m$ correlates with the discounted future extrinsic return. For a trajectory of length T, the discounted extrinsic return from time step t is

$$G_t^E = \sum_{k=t}^{T-1} \gamma^{k-t} R_k^E. \tag{3.8}$$

Within a minibatch with empirical means $\mu_{\tilde{I}}, \mu_G$ and standard deviations $\sigma_{\tilde{I}}, \sigma_G$, both the scaled intrinsic reward and the extrinsic return are standardized:

$$\hat{I}_t^m = \frac{\tilde{I}_t^m - \mu_{\tilde{I}}}{\sigma_{\tilde{I}}}, \qquad \hat{G}_t = \frac{G_t^E - \mu_G}{\sigma_G}. \tag{3.9}$$

The correlation-based loss is then

$$L_{\text{corr}}(\psi) = -\mathbb{E}\left[\hat{I}_t^m \hat{G}_t\right], \tag{3.10}$$

and to prevent degenerate scaling a log-space regularizer pulls β_ψ toward a reference value $\beta_0 = 1$:

$$L_{\text{reg}}(\psi) = \mathbb{E}\left[\left(\log \beta_\psi(s_t) - \log \beta_0\right)^2\right]. \tag{3.11}$$

The complete objective minimized at each update is

$$L_\beta(\psi) = L_{\text{corr}}(\psi) + \lambda_{\text{reg}} L_{\text{reg}}(\psi). \tag{3.12}$$

The correlation loss has a precise statistical interpretation, and its gradient has a precise sign condition.

Proposition 3.1 (Pearson Equivalence)**.** *In the limit of an infinitely large minibatch, the empirical estimator of $L_{corr}(\psi)$ in Equation* (3.10) *converges to the negative population Pearson correlation between $\tilde{I}_t^m$ and G_t^E:*

$$L_{corr}(\psi) = -\rho\left(\tilde{I}_t^m, G_t^E\right).$$

Proof in Appendix A.1.

Proposition 3.1 clarifies that minimizing L_{corr} is not a heuristic surrogate but is equivalent (up to a sign) to maximizing the Pearson correlation between the scaled intrinsic reward and the future extrinsic return. The next result shows that the gradient with respect to β_ψ at a state s has a clean directional interpretation.

Proposition 3.2 (Gradient Direction)**.** *Assume the trajectories used to estimate $L_{corr}(\psi)$ are sufficiently diverse so that perturbations of β_ψ at an individual state* *s have a negligible effect on the batch statistics $\mu_{\tilde{I}}, \sigma_{\tilde{I}}$. Then for every state $s \in \mathcal{S}$ that is visited with positive probability,*

$$\frac{\partial L_{corr}}{\partial \beta_\psi(s)} < 0 \quad \Longleftrightarrow \quad \mathbb{E}\big[I_t^{+,m}\, G_t^E \,\big|\, s_t = s\big] > \mathbb{E}\big[I_t^{+,m}\big]\, \mathbb{E}\big[G_t^E\big]\,.$$

Proof in Appendix A.2.

Proposition 3.2 formalizes the central design intuition of CARE: a gradient step on the correlation loss pushes $\beta_\psi(s)$ upward exactly at those states where intrinsic reward and future extrinsic return are positively correlated, and downward elsewhere. In other words, the Beta Network is supervised by the empirical correlation pattern of the agent's own experience.

3.3.3 First-Order Optimization vs. Meta-Gradient

A natural alternative to the correlation-based objective is to learn the intrinsic reward weighting via meta-gradients, as in Learning Intrinsic Rewards for Policy Gradient methods (LIRPG) [19]. In that approach, the parameters of the intrinsic reward function are updated so that, after a policy update using the shaped reward, the resulting policy maximizes the extrinsic objective. This requires differentiating through one or more inner policy updates, which is computationally expensive and can be unstable in deep reinforcement learning settings.

CARE replaces the meta-gradient by a first-order objective that supervises β_ψ directly using observed correlation. The cost difference is quantified by the following result.

Proposition 3.3 (Computational Cost)**.** *Per Beta-network update of size T, let K denote the number of inner policy update steps used in a meta-gradient estimator, $|\theta|$ the number of policy parameters, and $|\psi|$ the number of Beta-network parameters. Then a meta-gradient training step has worst-case time complexity $O(K\,T\,|\theta|)$, while a CARE first-order training step has time complexity $O(T\,|\psi|)$. In particular, when $|\psi| \ll |\theta|$ and $K \geq 1$, the CARE update is asymptotically cheaper by a factor of $K\,|\theta|/|\psi|$. Proof in Appendix A.4.*

In the implemented system, $|\psi|$ is roughly two orders of magnitude smaller than $|\theta|$, so the cost reduction is substantial. The price of this efficiency is a weaker form of supervision: instead of optimizing β_ψ directly for downstream extrinsic performance, CARE optimizes it for an empirical correlation that is a proxy for that performance. The next result describes the limit in which this proxy becomes uninformative.

Corollary 3.1 (Degeneration to Fixed Coefficient). *Consider the objective $L_\beta(\psi)$ in Equation (3.12). As $\lambda_{reg} \to \infty$, every minimizer $\psi^\star$ satisfies $\beta_{\psi^\star}(s) \to \beta_0 = 1$ uniformly in s, and the shaped reward in Equation (3.2) reduces to the fixed-coefficient form $\bar{r}_t = R_t^E + \alpha\, I_t^{+,m}$. Proof in Appendix A.5.*

Corollary 3.1 has both a theoretical and a practical reading. Theoretically, it shows that CARE strictly contains fixed-coefficient PPO+ICM as a limiting case, so it cannot perform worse than that baseline once λ_{reg} is tuned appropriately. Practically, it explains an empirically observed phenomenon: when extrinsic rewards are extremely sparse, L_{corr} provides little signal, the regularizer dominates, and CARE behaves like a fixed-coefficient method instead of degrading the policy.

3.4 Algorithm

This section formalizes the data structures, loss functions, and training procedure used to optimize the system end to end.

3.4.1 Rollout Buffer

Each PPO update operates on an on-policy rollout buffer of length T collected by the current policy. A single transition is the tuple

$$\xi_t \;=\; \big(s_t,\; a_t,\; \log \pi_\theta(a_t \mid s_t),\; R_t^E,\; s_{t+1},\; d_t\big), \tag{3.13}$$

where $d_t \in \{0,1\}$ is the episode termination flag. The buffer of length T is stored as the tensors

$$\mathcal{S}_{\text{tensor}} \in \mathbb{R}^{T\times d_s}, \quad \mathcal{A}_{\text{tensor}} \in \mathbb{N}^T, \quad \mathcal{R}_{\text{tensor}} \in \mathbb{R}^T, \quad \mathcal{S}'_{\text{tensor}} \in \mathbb{R}^{T\times d_s}, \quad \mathcal{D}_{\text{tensor}} \in \{0,1\}^T,$$

together with the log-probabilities $\mathcal{P}_{\text{tensor}} \in \mathbb{R}^T$. Following the implementation defaults, $T = 4 \times \text{max\_ep\_len}$ steps are collected before each PPO update.

3.4.2 Loss Functions

Four distinct losses are minimized at each update cycle. They share the same rollout buffer but are optimized with respect to disjoint parameter groups, so no joint optimization is required.

PPO Actor Loss $L^{\mathbf{PPO}}(\theta)$. Define the importance ratio $\rho_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ and let $\hat{A}_t$ denote the advantage estimate computed by Generalized Advantage Estimation (GAE) [20] from the shaped reward $\bar{r}_t$. The clipped surrogate objective [21] is

$$L^{\text{PPO}}(\theta) \;=\; -\,\mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\,\hat{A}_t,\;\text{clip}\big(\rho_t(\theta),\,1-\epsilon,\,1+\epsilon\big)\,\hat{A}_t\Big)\Big]. \tag{3.14}$$

Critic Loss $L^V(\mu)$. The value network V_μ is fitted by mean-squared error against the GAE return target $\hat{V}_t$:

$$L^V(\mu) \;=\; \mathbb{E}_t\Big[\big(V_\mu(s_t) - \hat{V}_t\big)^2\Big]. \tag{3.15}$$

ICM Loss $L_{\mathbf{ICM}}(\eta)$. As specified in Equation (3.3).

Beta Loss $L_\beta(\psi)$. As specified in Equation (3.12).

The four losses correspond to four parameter groups: θ (actor), μ (critic), η (ICM), and ψ (Beta Network). Gradients flow through each loss with respect to its own parameter group only; in particular, the Beta Network is not updated through L^{PPO} and the policy is not updated through L_β.

3.4.3 Training Procedure

Algorithm 1 specifies the complete CARE training loop. One iteration of the outer loop corresponds to a single PPO update cycle.

Algorithm 1 CARE training loop with PPO and ICM.

1: Initialize policy θ, critic μ, ICM η, Beta Network ψ
2: **for** update $= 1, 2, \ldots, U$ **do**
3:     Collect T on-policy transitions $\{\xi_t\}_{t=0}^{T-1}$ using π_θ
4:     Compute raw intrinsic rewards I_t via Equation (3.4)
5:     Standardize and rectify to obtain I_t^+ via Equation (3.5)
6:     Compute discounted extrinsic returns G_t^E via Equation (3.8)
7:     Update ICM parameters: $\eta \leftarrow \eta - \eta_{\text{lr}} \nabla_\eta L_{\text{ICM}}(\eta)$
8:     Update Beta Network: $\psi \leftarrow \psi - \psi_{\text{lr}} \nabla_\psi L_\beta(\psi)$
9:     Compute shaped rewards $\bar{r}_t = R_t^E + \alpha\, \beta_\psi(s_t)\, I_t^+$
10:     Compute GAE advantages $\hat{A}_t$ from $\{\bar{r}_t\}$ and V_μ
11:     **for** epoch $= 1, 2, \ldots, K$ **do**
12:         Update actor: $\theta \leftarrow \theta - \theta_{\text{lr}} \nabla_\theta L^{\text{PPO}}(\theta)$
13:         Update critic: $\mu \leftarrow \mu - \mu_{\text{lr}} \nabla_\mu L^V(\mu)$
14:     **end for**
15: **end for**

The algorithm preserves the standard PPO update structure (lines 11–14) and inserts only two additional optimization steps (lines 7–8) before the shaped reward is computed. This modularity is important: the same template applies if ICM is replaced by a count-based bonus or by RND, with only Equations (3.4)–(3.5) being substituted by the corresponding module-specific computation. Concrete numerical values for all hyperparameters and network architectures used in the experiments are listed in Appendix B.

Chapter 4

Experimental Setup and Results

4.1 Benchmark Environments

The empirical evaluation is conducted on five sparse-reward MiniGrid environments [15]. These environments are chosen because they represent different forms of exploration difficulty while maintaining a common gridworld interface.

Bảng 4.1: MiniGrid environments used in the experiments

Environment	Purpose in the evaluation
DoorKey-8x8	Tests whether the agent can discover a key, unlock a door, and shift from exploration to goal-directed behavior.
Empty-16x16	Represents an extreme sparse-reward setting with almost no intermediate feedback before the goal is reached.
RedBlueDoors-8x8	Requires opening two doors in the correct order, testing whether the method can suppress irrelevant exploration.
UnlockPickup	Extends DoorKey with an additional object manipulation stage, creating a hierarchical sparse-reward task.
KeyCorridorS3R3	A longer-horizon task with multiple rooms, keys, and locked doors, used to assess sustained exploration ability.

These tasks share several useful properties: sparse terminal rewards, egocentric partial observations, step penalties that discourage random motion, and stochastic initialization over multiple random seeds. Together, they form a suitable test bed for evaluating whether adaptive intrinsic reward scaling improves both efficiency and robustness.

4.2 Baselines and Evaluation Criteria

The experimental framing of this thesis treats CARE as an adaptive layer that can be placed on top of representative intrinsic reward modules. The three module families emphasized throughout the thesis are count-based bonuses, ICM, and RIDE. In the present implementation, the reported quantitative experiments instantiate the ICM branch of this framing, because it provides a controlled and computationally manageable case for measuring the effect of adaptive scaling.

Under that implementation, the proposed CARE method is compared against two categories of baselines:

- PPO trained only with extrinsic rewards;
- PPO combined with ICM using fixed intrinsic coefficients $\beta \in \{0.1, 0.2, 0.5, 1, 2\}$.

This comparison is important because it isolates the contribution of adaptive scaling from the contribution of intrinsic motivation itself. PPO provides a lower bound on exploration ability, while fixed-coefficient PPO+ICM shows how sensitive performance is to manual tuning. From the perspective of the broader thesis framing, the comparison also serves as the primary case study for understanding how CARE affects an intrinsic reward module once the module's raw signal is held fixed.

The evaluation focuses on four aspects:

- episode return over training time;
- sample efficiency in the early and middle stages of learning;
- variance across random seeds;
- qualitative behavior of the learned scaling coefficient $\beta(s)$ and what it reveals about module-dependent exploration control.

4.3 Experimental Configuration

All reported runs share the same PPO backbone and ICM architecture so that the comparison is fair. The adapted paper fixes the global intrinsic coefficient at $\alpha = 0.001$ and clips the learned Beta Network output to $[0.1, 2.0]$. The Beta Network uses a hidden representation size of 256 with encoder depth 2. The regularization weight for the correlation objective is $\lambda_{\text{reg}} = 10^{-3}$, the reference value is $\beta_0 = 1.0$, gradient clipping is set to 1.0, and weight decay is 10^{-6}.

Although the current implementation focuses on ICM, the same CARE formulation would be applied to count-based and RIDE modules by replacing the raw intrinsic reward source while keeping the state-dependent scaling and correlation objective unchanged. This is why the chapter is written not only as a report of one implementation, but also as an empirical framing for studying CARE across intrinsic reward modules.

Bảng 4.2: Key hyperparameters reported for CARE

Hyperparameter	**Value**
Intrinsic strength α	0.001
Beta output bounds	$[0.1, 2.0]$
Encoder size	256
Encoder depth	2
Regularization weight λ_{reg}	10^{-3}
Reference value β_0	1.0
Gradient clipping	1.0
Weight decay	10^{-6}

The results are reported over five random seeds. This setup is sufficient to assess whether adaptive scaling consistently improves learning behavior rather than benefiting from a single favorable initialization.

4.4 Results

The quantitative experiments reported in this thesis correspond to the ICM instantiation of CARE. Within that setting, the results show several consistent trends. First, CARE performs strongly across environments with sparse but still informative extrinsic rewards. In DoorKey-8x8, RedBlueDoors-8x8, UnlockPickup, and KeyCorridorS3R3, the adaptive approach achieves faster learning and lower variance than most fixed-coefficient baselines. This result indicates that CARE improves sample efficiency and makes exploration behavior more stable across runs.

Second, the comparison against fixed-coefficient PPO+ICM highlights the difficulty of manual tuning. A coefficient that works well in one environment can be too small or too large in another. Small values underemphasize exploration and slow down learning, while large values can inject excessive noise and interfere with exploitation. CARE mitigates both problems by learning to amplify intrinsic rewards only in states where they correlate with future extrinsic success. In the broader module-level framing of this thesis, this result supports the hypothesis that CARE is most useful when an intrinsic reward signal is informative but unevenly relevant across the state space.

Third, the method exhibits a natural transition from exploration to exploitation. As training progresses and the policy begins to discover task-relevant behaviors, the effective intrinsic contribution tends to decrease. This suggests that CARE does not merely add curiosity-driven noise, but instead learns to reduce exploration pressure once the task structure becomes clearer.

4.5 Analysis of the Learned Scaling Behavior

The paper reports that the learned distribution of $\beta(s)$ evolves differently across environments. In DoorKey-8x8 and RedBlueDoors-8x8, the distribution is initially concentrated near its reference value, then gradually broadens and becomes more structured. This pattern suggests that the Beta Network learns to partition the state space into regions where different exploration intensities are appropriate. Toward the end of training, the distribution shifts toward lower values, reflecting the decreasing marginal value of intrinsic rewards once the policy has matured.

In contrast, Empty-16x16 behaves differently. Because extrinsic rewards remain almost zero for most trajectories, the correlation objective receives very weak supervision. As a result, the Beta Network stays close to its initialization and exhibits limited adaptation. Importantly, this failure mode is graceful rather than unstable: the method effectively defaults to a nearly fixed intrinsic coefficient instead of diverging. This observation confirms both a strength and a limitation of CARE. The method is robust when correlation signals are weak, but meaningful adaptation requires at least occasional extrinsic feedback. From a module-level perspective, this also suggests that CARE will be most effective for intrinsic reward modules whose signals can be related to eventual task progress often enough for the correlation objective to learn from experience.

4.6 Discussion

Overall, the empirical evidence supports the main claim of the thesis: state-dependent intrinsic reward scaling can improve sparse-reward reinforcement learning when the environment provides enough structure for the agent to relate exploration to eventual success. In the implemented ICM case study, CARE is especially beneficial in environments where exploration needs change across states and over time. The method is less effective in extremely sparse tasks where the extrinsic return is too weak to supervise adaptation. Therefore, CARE should be viewed as a practical middle ground between fixed intrinsic scaling and more expensive meta-learning approaches, and as a promising adaptive wrapper for broader module families such as count-based bonuses and RIDE.

Conclusion and Future Work

This thesis studies Correlation-Aware Reward Exploration (CARE) in reinforcement learning. Instead of treating CARE as a method used with only one type of curiosity reward, this work views it as an adaptive layer to study how state-dependent scaling of intrinsic rewards affects learning. The thesis considers three main types of intrinsic rewards: count-based bonuses, ICM, and RND. CARE is used as the mechanism that decides how much these rewards should influence learning in each state. By matching the scaled intrinsic reward with the discounted future extrinsic return, CARE offers a simple way to balance exploration and exploitation during training without using costly meta-gradient methods.

Experiments on MiniGrid use PPO+ICM as the main case study and show that CARE improves learning stability and sample efficiency in several sparse-reward tasks, especially when extrinsic rewards are rare but still useful. Compared to fixed coefficients that need manual tuning, the adaptive approach works more consistently across different environments and random seeds. However, the results also show a limitation: when extrinsic rewards are almost always missing, the correlation signal becomes weak, and the scaling factor behaves like a fixed coefficient. This suggests that the effectiveness of CARE depends on how useful the intrinsic reward is and how often it relates to future task success.

There are several directions for future work. First, the experiments can be extended by testing CARE more fully on count-based bonuses and RND, not only ICM. Second, the method can be applied to broader settings such as continuous control, multi-task RL, and embodied navigation. Third, a clearer theoretical analysis of the correlation objective could help explain when adaptive scaling is most useful across different intrinsic reward types. These directions can further improve the use of adaptive intrinsic rewards in reinforcement learning.

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Chapter A

Mathematical Proofs

This appendix collects the full proofs of the formal results stated in Chapter 3. Each section below corresponds to a single proposition, lemma, or corollary, and may be read independently of the others. Throughout the appendix, expectations are taken with respect to the on-policy distribution induced by the current policy π_θ unless stated otherwise; standard regularity conditions (existence of first and second moments, measurability of β_ψ) are assumed.

A.1 Proof of Proposition 3.1: Correlation Equivalence

Chứng minh. Let $\tilde{I}_t \equiv \tilde{I}_t^m$ denote the scaled intrinsic reward and let $G_t \equiv G_t^E$ denote the discounted extrinsic return. Both random variables are assumed to have finite, nonzero variance under the on-policy distribution. From Equations (3.9) and (3.10),

$$L_{\text{corr}}(\psi) \;=\; -\,\mathbb{E}\big[\hat{I}_t\,\hat{G}_t\big] \;=\; -\,\mathbb{E}\left[\frac{\tilde{I}_t - \mu_{\tilde{I}}}{\sigma_{\tilde{I}}}\cdot\frac{G_t - \mu_G}{\sigma_G}\right].$$

Pulling the constants $\sigma_{\tilde{I}}$ and σ_G out of the expectation, which is valid because they are non-random in the population limit,

$$L_{\text{corr}}(\psi) \;=\; -\,\frac{1}{\sigma_{\tilde{I}}\,\sigma_G}\,\mathbb{E}\big[\big(\tilde{I}_t - \mu_{\tilde{I}}\big)\big(G_t - \mu_G\big)\big] \;=\; -\,\frac{\mathrm{Cov}(\tilde{I}_t,\,G_t)}{\sigma_{\tilde{I}}\,\sigma_G}.$$

The right-hand side is precisely the negative Pearson correlation coefficient,

$$L_{\text{corr}}(\psi) \;=\; -\,\rho\big(\tilde{I}_t,\,G_t\big)\,.$$

In a finite minibatch, the empirical estimator $\widehat{L}_{\text{corr}}$ replaces $\mu_{\tilde{I}}, \mu_G, \sigma_{\tilde{I}}, \sigma_G$ with sample analogues. By the strong law of large numbers and the continuous mapping theorem applied to the ratio of sample covariance and sample standard deviations, $\widehat{L}_{\text{corr}} \xrightarrow{\text{a.s.}} -\rho(\tilde{I}_t, G_t)$ as the minibatch size tends to infinity, which establishes the claim. □

A.2 Proof of Proposition 3.2: Gradient Direction

Chứng minh. Fix a state $s \in \mathcal{S}$ visited with positive probability, and consider an infinitesimal perturbation of β_ψ at s. By assumption, this perturbation has negligible effect on the batch statistics $\mu_{\tilde{I}}$ and $\sigma_{\tilde{I}}$, so for the purpose of computing the partial derivative they may be treated as constants. The same applies to μ_G and σ_G, which do not depend on ψ at all.

Recall that $\tilde{I}_t = \beta_\psi(s_t)\, I_t^+$, where we write $I_t^+ \equiv I_t^{+,m}$ for brevity. Differentiating Equation (3.10) with respect to the value of β_ψ at the specific state s,

$$\begin{aligned}\frac{\partial L_{\text{corr}}}{\partial \beta_\psi(s)} &= -\frac{\partial}{\partial \beta_\psi(s)}\,\mathbb{E}\left[\frac{\tilde{I}_t - \mu_{\tilde{I}}}{\sigma_{\tilde{I}}} \cdot \hat{G}_t\right] \\ &= -\frac{1}{\sigma_{\tilde{I}}}\,\mathbb{E}\big[\mathbf{1}\{s_t = s\}\, I_t^+\, \hat{G}_t\big] \\ &= -\frac{\mathbb{P}(s_t = s)}{\sigma_{\tilde{I}}}\,\mathbb{E}\big[I_t^+\, \hat{G}_t \,\big|\, s_t = s\big]\,.\end{aligned}$$

Using $\hat{G}_t = (G_t - \mu_G)/\sigma_G$ and the fact that $\mu_G = \mathbb{E}[G_t]$ does not depend on conditioning on $s_t = s$ within the standardization step,

$$\mathbb{E}\big[I_t^+\, \hat{G}_t \,\big|\, s_t = s\big] = \frac{1}{\sigma_G}\Big(\mathbb{E}[I_t^+\, G_t \mid s_t = s] - \mu_G\,\mathbb{E}[I_t^+ \mid s_t = s]\Big).$$

Therefore, since $\sigma_{\tilde{I}}, \sigma_G, \mathbb{P}(s_t = s) > 0$, the sign of $\partial L_{\text{corr}}/\partial \beta_\psi(s)$ is opposite to the sign of

$$\Delta(s) \coloneqq \mathbb{E}[I_t^+\, G_t \mid s_t = s] - \mathbb{E}[G_t]\,\mathbb{E}[I_t^+ \mid s_t = s].$$

Under the additional simplifying assumption that the conditional and marginal distributions of I_t^+ have approximately the same mean (which holds in the large-batch limit when s is one of many visited states), $\mathbb{E}[I_t^+ \mid s_t = s] \approx \mathbb{E}[I_t^+]$, and the criterion reduces to

$$\frac{\partial L_{\text{corr}}}{\partial \beta_\psi(s)} < 0 \iff \mathbb{E}[I_t^+\, G_t \mid s_t = s] > \mathbb{E}[I_t^+]\,\mathbb{E}[G_t].$$

The right-hand side is exactly the conditional positive-correlation condition stated in Proposition 3.2. □

A.3 Proof of Lemma 3.1: Bounded Shaped Reward

Chứng minh. By Equation (3.2),

$$\bar{r}_t - R_t^E = \alpha\, \beta_\psi(s_t)\, I_t^{+,m}.$$

Taking absolute values and using $\alpha > 0$,

$$\left|\bar{r}_t - R_t^E\right| \;=\; \alpha \,\left|\beta_\psi(s_t)\right|\, \left|I_t^{+,m}\right|.$$

By Definition 3.4, $\beta_\psi(s_t) \in [\beta_{\min}, \beta_{\max}]$ with $\beta_{\min} > 0$, so $|\beta_\psi(s_t)| \leq \beta_{\max}$. By assumption, $|I_t^{+,m}| \leq M$ almost surely. Combining these bounds,

$$\left|\bar{r}_t - R_t^E\right| \;\leq\; \alpha\, \beta_{\max}\, M,$$

which establishes the claim. □

A.4 Proof of Proposition 3.3: Computational Cost Comparison

Chứng minh. We measure cost in scalar floating-point operations and ignore lower-order terms.

Meta-gradient training step. A meta-gradient training step for a parameterized intrinsic reward function (LIRPG [19]) consists of the following operations on a buffer of T transitions: (i) compute the policy gradient at the current parameters and apply K inner update steps to obtain a perturbed policy θ', each of which costs $O(T\,|\theta|)$ to evaluate the policy gradient and $O(|\theta|)$ to apply the update; and (ii) backpropagate the meta-objective evaluated at θ' through the chain of K inner updates to obtain a gradient with respect to the intrinsic reward parameters. The backward pass through K unrolled updates costs $O(K\,T\,|\theta|)$ in time and $O(K\,|\theta|)$ in memory because each inner step must be cached. Summing the forward and backward costs, the worst-case time complexity per buffer update is

$$C_{\text{meta}} \;=\; O(K\,T\,|\theta|).$$

CARE training step. A CARE training step for the Beta Network on the same buffer of T transitions consists of: (i) one forward pass through β_ψ for each of the T states, at total cost $O(T\,|\psi|)$; (ii) computation of $\tilde{I}_t$ and $\hat{I}_t$, $\hat{G}_t$, all of which take $O(T)$ time; and (iii) one backward pass through β_ψ to compute $\nabla_\psi L_\beta$, at cost $O(T\,|\psi|)$. The total time complexity per buffer update is therefore

$$C_{\text{CARE}} \;=\; O(T\,|\psi|).$$

Asymptotic ratio. Taking the ratio,

$$\frac{C_{\text{meta}}}{C_{\text{CARE}}} \;=\; \Theta\!\left(\frac{K\,|\theta|}{|\psi|}\right).$$

When $|\psi| \ll |\theta|$ and $K \geq 1$, this ratio is large, which is what the proposition asserts. □

A.5 Proof of Corollary 3.1: Degeneration to Fixed Coefficient

Chứng minh. Recall from Equation (3.12) that

$$L_\beta(\psi) \;=\; L_{\text{corr}}(\psi) \;+\; \lambda_{\text{reg}}\, L_{\text{reg}}(\psi),$$

with $L_{\text{reg}}(\psi) = \mathbb{E}\big[(\log \beta_\psi(s_t) - \log \beta_0)^2\big]$ and $\beta_0 = 1$. By Proposition 3.1, $L_{\text{corr}}(\psi) = -\rho(\tilde{I}_t, G_t) \in [-1, 1]$, so L_{corr} is uniformly bounded. The regularizer $L_{\text{reg}}(\psi)$ is non-negative and equals zero if and only if $\beta_\psi(s) = 1$ almost everywhere with respect to the state-visitation distribution.

Let $\psi^\star_\lambda$ denote any minimizer of L_β with regularization weight $\lambda \equiv \lambda_{\text{reg}}$. By optimality,

$$L_{\text{corr}}(\psi^\star_\lambda) \;+\; \lambda\, L_{\text{reg}}(\psi^\star_\lambda) \;\leq\; L_{\text{corr}}(\psi_0) \;+\; \lambda\, L_{\text{reg}}(\psi_0),$$

where ψ_0 is any reference parameter satisfying $\beta_{\psi_0}(s) \equiv 1$, for which $L_{\text{reg}}(\psi_0) = 0$. Rearranging and using the bound $|L_{\text{corr}}| \leq 1$,

$$\lambda\, L_{\text{reg}}(\psi^\star_\lambda) \;\leq\; L_{\text{corr}}(\psi_0) - L_{\text{corr}}(\psi^\star_\lambda) \;\leq\; 2.$$

Dividing by $\lambda > 0$,

$$L_{\text{reg}}(\psi^\star_\lambda) \;\leq\; \frac{2}{\lambda} \;\xrightarrow[\lambda\to\infty]{}\; 0.$$

Since $L_{\text{reg}}(\psi^\star_\lambda) \to 0$, the squared deviation $(\log \beta_{\psi^\star_\lambda}(s_t) - 0)^2$ has expectation tending to zero. Therefore $\log \beta_{\psi^\star_\lambda}(s_t) \to 0$ in L^2 under the state-visitation distribution, which implies $\beta_{\psi^\star_\lambda}(s) \to 1$ in L^2 on the support of that distribution.

Substituting $\beta_\psi(s) = 1$ into the shaped reward of Equation (3.2) yields

$$\bar{r}_t \;=\; R^E_t \;+\; \alpha \cdot 1 \cdot I^{+,m}_t \;=\; R^E_t \;+\; \alpha\, I^{+,m}_t,$$

which is exactly the fixed-coefficient PPO+ICM update with intrinsic strength α. □

Chapter B

Implementation Details

This appendix records the network architectures, hyperparameters, and training schedule used in the experiments reported in Chapter 3 and Chapter 4. All values reflect the canonical settings adopted across the six benchmark MiniGrid environments and are extracted from the project source code.

B.1 Network Architectures

Bảng B.1: Beta Network β_ψ architecture.

Component	**Specification**
Input	state s_t flattened to $\mathbb{R}^{d_s}$
Encoder layer 1	Linear($d_s \to 256$), Tanh
Encoder layer 2	Linear($256 \to 256$), Tanh
Head layer 1	Linear($256 \to 128$), Tanh
Head output	Linear($128 \to 1$) producing $\log \beta_\psi(s_t)$
Output bound	$\beta_\psi(s_t) = \exp\big(\text{clamp}(\log \beta_\psi, \log \beta_{\text{min}}, \log \beta_{\text{max}})\big)$
$[\beta_{\text{min}}, \beta_{\text{max}}]$	[0.1, 2.0]
Encoder weight init	$\mathcal{N}(0, 1/\sqrt{d_{\text{in}}})$
Output bias init	$\log(1.0) = 0$ (so $\beta_\psi \approx 1$ at initialization)

Bảng B.2: Intrinsic Curiosity Module (ICM) architecture.

Component	**Specification**
Encoder ϕ layer 1	Linear($d_s \to 256$), Tanh
Encoder ϕ layer 2	Linear($256 \to 256$), Tanh
Latent dimension	256
Forward model f_η^F	Linear($256 + \lvert\mathcal{A}\rvert \to 256$), Tanh, Linear($256 \to 256$)
Inverse model f_η^I	Linear($512 \to 256$), Tanh, Linear($256 \to \lvert\mathcal{A}\rvert$)
Forward weight α_F in L_{ICM}	0.2
Inverse weight α_I in L_{ICM}	0.8
Action representation in forward model	one-hot $\in \mathbb{R}^{\lvert\mathcal{A}\rvert}$
Encoder gradient flow	inverse loss only (forward loss detached)

Bảng B.3: Actor and critic network architectures used by PPO.

Component	**Specification**
Actor body	Linear($d_s \to 64$), Tanh, Linear($64 \to 64$), Tanh
Actor head	Linear($64 \to \lvert\mathcal{A}\rvert$) producing logits; sampling via Categorical(logits)
Critic body	Linear($d_s \to 64$), Tanh, Linear($64 \to 64$), Tanh
Critic head	Linear($64 \to 1$)

B.2 Hyperparameters

Bảng B.4: PPO hyperparameters.

Symbol	**Description**	**Value**
K	Number of PPO gradient epochs per update	80
ϵ	PPO clip parameter	0.2
γ	Discount factor	0.99
λ	GAE bias-variance trade-off	0.95
θ_{lr}	Actor learning rate	3×10^{-4}
μ_{lr}	Critic learning rate	1×10^{-3}
Entropy coefficient	Weight on $-H[\pi_\theta]$ in PPO loss	0.01
Value loss coefficient	Weight on critic loss in joint update	0.5

Bảng B.5: ICM hyperparameters.

Symbol	**Description**	**Value**
η_{lr}	ICM learning rate	1×10^{-3}
ICM epochs	Number of gradient epochs per buffer update	4
ICM batch size	Mini-batch size within each epoch	64

Bảng B.6: CARE hyperparameters and intrinsic reward strengths.

Symbol	**Description**	**Value**
α (ICM)	Global intrinsic strength for ICM	1×10^{-3}
α (count)	Global intrinsic strength for count-based	1×10^{-3}
α (RND)	Global intrinsic strength for RND	1×10^{-3}
β_{min}	Lower bound on Beta Network output	0.1
β_{max}	Upper bound on Beta Network output	2.0
β_0	Reference value for log-space regularization	1.0
λ_{reg}	Regularization weight in L_β	1×10^{-3}
ψ_{lr}	Beta Network learning rate	5×10^{-4}
Weight decay (ψ)	L_2 weight decay on Beta Network	1×10^{-6}
Gradient clipping (ψ)	Max L_2 norm on $\nabla_\psi L_\beta$	1.0
ε	Numerical stability constant in standardization	1×10^{-8}

Bảng B.7: Other count-based and RND hyperparameters.

Symbol	**Description**	**Value**
d_{hash}	Hash dimension for count-based exploration	32
Bonus type (count)	Form of count-based bonus	$1/\sqrt{N(s)}$
RND learning rate	Predictor network learning rate	1×10^{-3}
RND epochs	Number of gradient epochs per buffer update	4
RND batch size	Mini-batch size within each epoch	64

Bảng B.8: Training schedule.

Symbol	**Description**	**Value**
max_ep_len	Maximum steps per episode	1,000
T	Rollout buffer length (PPO update period)	4,000
max_steps	Total environment interaction steps per seed	1,000,000
U	Total number of PPO updates per seed	250
seeds	Independent training seeds per configuration	$\{1, 2, 3, 4, 5\}$
Checkpoint interval	Frequency of model and trajectory saving	every 100,000 steps

The values listed in Tables B.1–B.8 are the canonical settings adopted across all benchmark MiniGrid environments. Environment-specific overrides, when present, are reported in Chapter 4.